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Microstructures, mechanical properties and deformation of near-rapidly solidified low-density Fe-20Mn-9Al-1.2C-xCr steels

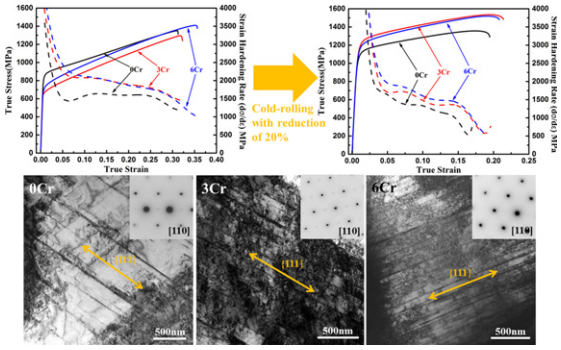
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HIGHLIGHTS

- Cr addition increases δ-ferrite content and decreases nano-scale κ-carbide content in low-density Fe-20Mn-9Al-1.2C steel.
- The decrease of nano-scale κ-carbide promotes generation of the slip bands and increases its strain hardening rate.
- A double (Slip-band and Microband) hardening mechanism is proposed for strain hardening of the low-density steel strips.

GRAPHICAL ABSTRACT



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ABSTRACT

The effects of Cr addition on microstructures, mechanical properties and deformation behavior of near-rapidly solidified low-density Fe-20Mn-9Al-1.2C (wt. %) steel strips were investigated. It was found that Cr addition increased the volume fraction and size of δ-ferrite, and decreased the precipitation of κ-carbides in this low-density steel strip. The yield strength of the near-rapidly solidified steel strips was decreased significantly, but the ultimate tensile strength and the total elongation didn't obviously change. However, after cold rolling with 20% reduction, the yield strength and ultimate tensile strength of Cr-containing steel strips were significantly higher than that of the Cr-free steel strip. It was attributed to the fact that the decrease of κ-carbides caused by Cr addition facilitated the formation of high-density slip bands during cold rolling deformation. The ultimate tensile strength and total elongation of cold-rolled 3Cr (wt. %) steel strip were 1270 MPa and 21%, respectively. A double (Slip-band and Microband) hardening mechanism was proposed for the strain hardening behavior of the low-density steel strips, and the number of nanoscale κ-carbides played a crucial role in the strain hardening rate at the early strain stage.

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1. Introduction

Due to increasingly rigorous demands of energy conservation and emission reduction, the Fe-Mn-Al-C steel with the excellent mechanical properties and low density has extensive attention in the automotive field [1–4]. According to the constitution phase, Fe-Mn-Al-C steel is

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Table 1
Measured compositions of the studied steel strips in this work (wt. %).

Samples	Mn	Al	C	Cr	S	Fe
0Cr steel	20.43	9.16	1.17	0	0.0006	Bal.
3Cr steel	20.04	9.09	1.23	3.09	0.0009	Bal.
6Cr steel	20.18	9.12	1.20	6.08	0.0011	Bal.

mainly divided into four categories: fully ferritic steel, fully austenitic steel, ferrite based duplex steel and austenite based duplex steel [5]. Because of the effective solid solution and the precipitation hardening of nano-scale particles, superior mechanical properties are obtained in the austenite-based duplex steel [6]. This steel consists of the austenite matrix with the κ-carbide precipitation and a small quantity of ferrite, also known as TRIPLEX steel [3,6–8]. Due to the high stacking fault energy (SFE) caused by high Al content, shear band induced plasticity (SIP), microband induced plasticity (MBIP) and slip band refinement-induced plasticity (SRIP) are considered as the main deformation mechanisms in this steel [6,9–14].

The mechanical properties of the Fe-Mn-Al-C steel are usually improved by adding alloying elements such as Mo, Ni, Nb, Cr, etc. [15–18]. Studies showed that the addition of Cr could significantly improve the corrosion resistance [19–21], high temperature oxidation resistance [22], environmental brittleness [19] and the tear resistance [23] of the Fe-Mn-Al-C steel. In addition, the influence of Cr on the phase transformation process in the low-density Fe-Mn-Al-C steel has also received extensive attention as a key research object [24–28]. M.S. Chen et al. [25] found that a phase transition of $\alpha + B2 \rightarrow \alpha + B2 + DO3 \rightarrow \alpha + DO3$ occurred in Fe-30Mn-9Al-(0.6, 0.9 and 1.2)C-(3, 5 and 7)Cr (wt. %) alloy with increasing Cr content. Cr addition also expands the A12α-Mn and DO3 phase-field regions [27]. The results obtained by a CALPHAD method [29] showed that Cr addition made the γ/γ+κ and the γ/γ+α phase boundaries move to the high-C side and the γ+α two-phase region expanded to the high-Al side in the phase diagrams.

However, the mechanical properties and deformation behavior of the austenite-based duplex Fe-Mn-Al-C steel with addition of Cr element have received little attention. The present study is designed to determine the influence of Cr addition on the microstructures, mechanical properties and deformation behavior of austenite-based low-density Fe-20Mn-9Al-1.2C (wt. %) steel.

In this work, a near-net shape method was used to prepare low-density steel strips which experienced near-rapid solidification with a cooling rate of 1–1 × 10^4 K/s [30,31]. Due to the high cooling rate, near-rapid solidification has features of expanding the solid solubility limit, refining the microstructure and forming the metastable phase [30–33]. Moreover, the near-net shape method can directly produce thin steel strips with a thickness of several mm, which can reduce the energy consumption produced by rolling. Thus it is relatively simple and energy-saving. Previous study [30] has shown that the mechanical properties of as-cast steel strips prepared by this method could reach or exceed that of the steel strips prepared by a conventional method. However, cold rolling is a common manufacturing process for the steels [1,3,14,16], so the properties of cold-rolled steels are of particular concern. Therefore, the as-cast and cold-rolled steel strips were investigated in this work. More importantly, the solidification condition of the near-rapid solidification is close to that of thin strip continuous casting [34]. It also contributes to the application of this high-efficiency and energy-saving thin strip continuous casting.

2. Experimental

In this work, the Fe-20Mn-9Al-1.2C-xCr (in wt. %, x = 0, 3 and 6, the same as below) steel strips were prepared by a centrifugal casting under near-rapid solidification in an argon atmosphere. A 100g alloy ingot was prepared by melting 99.99% pure iron, 99.9% pure manganese, 99.9% pure aluminum, 99.9% pure chromium and 99.8% high-purity graphite by a cold-crucible levitation melting setup. The metal ingot was remelted by induction melting, and the molten metal was poured into a copper mold rotating at 600 rpm. The as-cast steel strips with size of 80 mm × 60 mm × 2.5 mm were obtained. Previous studies [30] have found that annealing at 600 °C for 3 h promoted the precipitation of κ-carbides without coarsening, contributing to obtain the better comprehensive mechanical properties of the low-density steel. Some as-cast steel strips were annealed at 600 °C for 3 h in a tubular vacuum furnace, which were referred as "annealed steel strips". Some as-cast steel strips were cold rolled (CR) with 20% reduction after stress relief annealing at 400 °C for 1 h, which was denoted by "CR20%", hereafter. The cold rolling with 20% reduction can eliminate some micropores caused by solidification shrinkage and avoid the great influence on the structure of the steel strips. The cold rolling was achieved by multipass rolling and carried out on a twin roll mill (Φ160). The CR20% steel strips were further annealed at 600 °C for 3 h, which was denoted by "CR20% + annealed", hereafter.

The measured average density of the 0Cr, 3Cr and 6Cr steel strips was 6.71 g/cm^3, 6.69 g/cm^3 and 6.66 g/cm^3, respectively, which was performed by an AccuPyc II 1340 Gas Displacement Density Analyzer. The compositions of the steel strips were measured by using an Inductively Coupled Plasma Mass Spectrometry (ICP) and a CS2800 carbon-sulfur analyzer. The measured compositions are shown in Table 1. For clarification, the low-density steel strips with 0, 3 wt % and 6 wt % Cr were referred as "0Cr", "3Cr" and "6Cr", respectively.

Phase characterization was performed by an X-ray Diffractometer (Rigaku, D/Max-2200, Cu target operated at 40 kV and 40 mA) with a graphite monochromator. The wavelength of the radiation was λ =

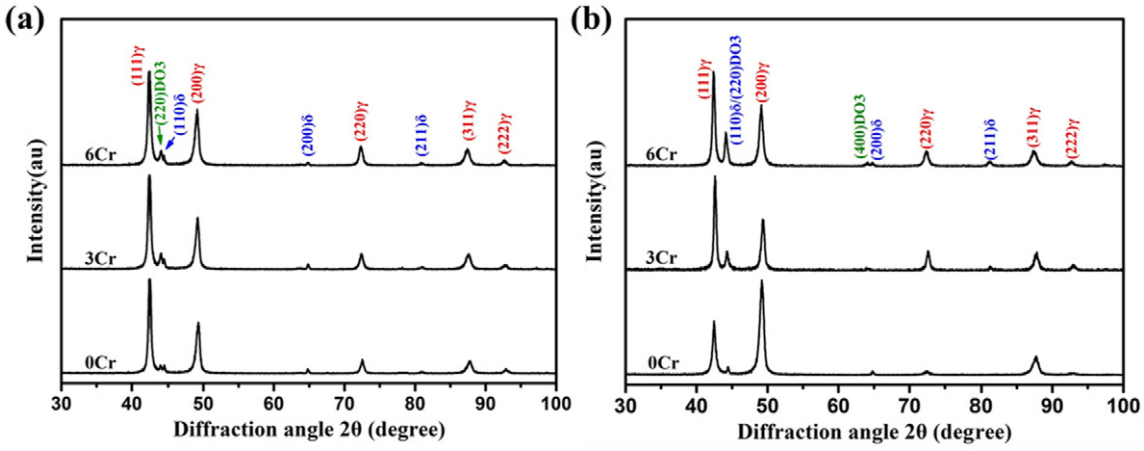


Fig. 1. X-ray diffraction patterns of the low-density Fe-20Mn-9Al-1.2C-xCr steel strips: (a) as-cast; (b) annealed.

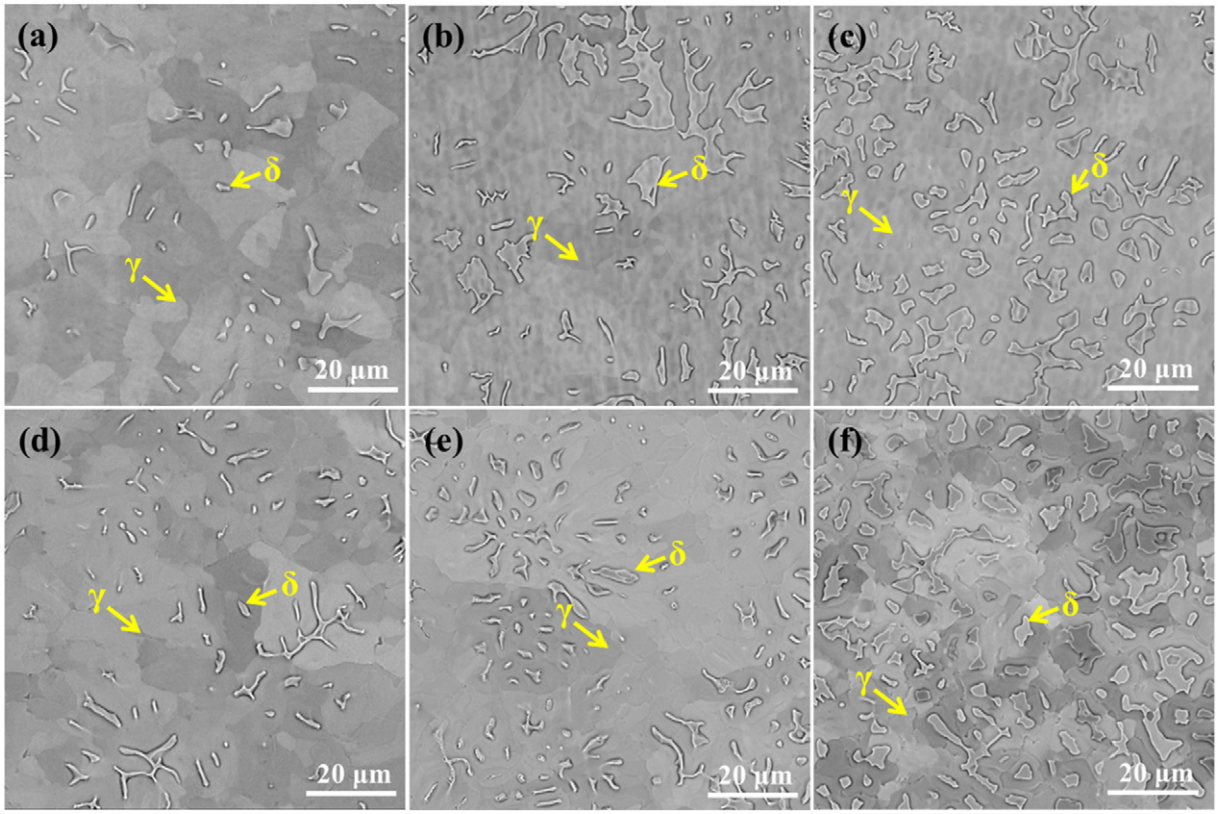


Fig. 2. SEM micrographs of the Fe-20Mn-9Al-1.2C-xCr steel strips. As-cast: (a) 0Cr, (b) 3Cr, (c) 6Cr; annealed: (d) 0Cr, (e) 3Cr, (f) 6Cr.

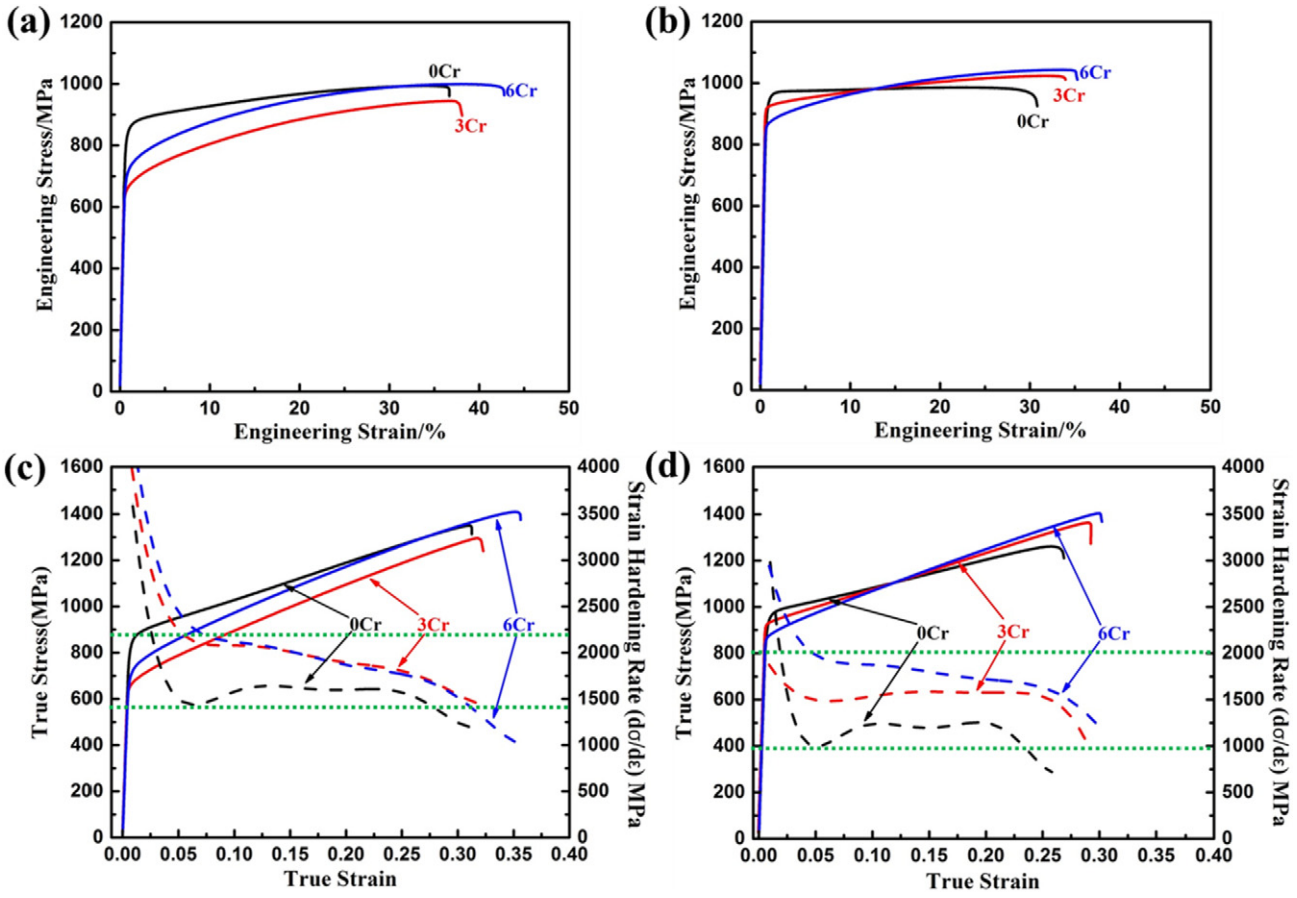


Fig. 3. Engineering stress-strain curves of the as-cast (a) and annealed (b) steel strips; true stress-strain curves and strain hardening rate curves of the as-cast (c) and annealed (d) steel strips.

Table 2
Mechanical properties of the as-cast and annealed steel strips.

Samples		YS/MPa	UTS/MPa	TE/%	UTS × TE/GPa %
0Cr	as-cast	836 ± 7	972 ± 22	35 ± 8	34.0
	annealed	967 ± 28	1016 ± 22	30 ± 1	30.5
3Cr	as-cast	642 ± 6	943 ± 10	38 ± 3	35.8
	annealed	919 ± 5	1033 ± 10	33 ± 4	34.1
6Cr	as-cast	718 ± 17	1001 ± 3	38 ± 4	38.0
	annealed	859 ± 7	1028 ± 15	32 ± 4	32.9

0.154056 nm and the scanning angle was 30°–100°, scanning mode was continuous scanning with a speed of 4°/min. The microstructure was characterized by a Scanning Electron Microscope (SEM, Phenom Pro.) after mechanical polishing and etching. The composition of the etching solutions was 3.3 vol % nitric acid, 3.3 vol % acetic acid and 0.1 vol % hydrofluoric acid. The etching time was 3 s. The SEM photographs were taken at a position of 0.5 mm from the strip surface along the thickness direction. The average phase fraction was measured by using an image processing software (Image-Pro Plus, IPP) in ten different positions. The specimens of transmission electron microscope (TEM) were prepared by using a twin-jet electrolytic polishing at −30 °C and 30 V, with a mixed solution of 10% perchloric acid and 90% alcohol. TEM was performed by using JEOL-2010F operated at 200 kV. Tensile tests at room temperature were performed at least 3 times by a MTS Criterion 44 Mechanical Property Tester. The tensile specimens with a gage width of 4 mm and a gage length of 12 mm were machined from the steel strips obtained under different conditions. The tensile testing was performed in accordance with Chinese Standard (GB/T-228.1-2010) at a constant strain rate of $2.5 \times 10^{-4}\ s^{-1}$.

3. Results

3.1. Microstructures and mechanical properties of steel strips

3.1.1. Microstructures of steel strips

Fig. 1 is X-ray diffraction patterns of the as-cast and annealed low-density Fe-20Mn-9Al-1.2C-xCr steel strips, which shows that the steel strips mainly consist of austenite and ferrite. The ferrite should be high temperature δ-ferrite reserved from high temperature due to the high cooling rate during near-rapid solidification [30]. There is a double-peak between 43.5° and 45.0° in the as-cast steel strips. According to the previous study [30], one is (110)δ and the other is identified as (220)DO3 with the space group of Fm3m [35]. The ordered DO3 phase is a body centered cubic crystal with a lattice parameter a = 0.565 nm in the ferrite [26], in which Fe or Mn atoms occupy cornered positions, and Al or Fe/Mn atoms occupy every other centered positions [35]. The area ratio of the (110)δ peak to (220)DO3 peak gradually decreases with the increase of Cr content. Thus, the Cr element promotes the transformation from δ-ferrite to the ordered DO3 phase. The double-peak between 43.5° and 45.0° changes into a single-peak when the steel strips are annealed at 600 °C for 3 h, as shown in Fig. 1 (b). It indicates that almost all the ferrite transforms into the ordered DO3 phase. Moreover, the intensity of the single-peak is significantly enhanced with increasing the Cr content in the annealed steel strips, which indicates that the content of the ordered DO3 phase does increase.

Fig. 2 shows the SEM micrographs of as-cast and annealed Fe-20Mn-9Al-1.2C-xCr steel strips. It shows that the irregularly shaped δ-ferrite distributes in the γ-austenite matrix, and the Cr addition makes the short rod-shape ferrite change into block ferrite. The volume fractions of δ-ferrite in the as-cast 0Cr, 3Cr and 6Cr steel strips are 3%, 12% and 18%, respectively. And, annealing at 600 °C for 3 h makes the volume fraction of the δ-ferrite in the 0Cr, 3Cr and 6Cr steel strips increase to 6%, 14% and 24%, respectively.

3.1.2. Mechanical properties of steel strips

Fig. 3 (a) and (b) show the engineering stress-strain curves of the as-cast and annealed Fe-20Mn-9Al-1.2C-xCr steel strips. A summary of the mechanical properties of the steel strips is listed in Table 2. It can be seen that the addition of Cr significantly decreases the yield strength (YS), but has no obvious effect on the elongation of the Fe-20Mn-9Al-1.2C steel strip. The annealing treatment increases the yield strength of 0Cr, 3Cr and 6Cr steel strips by 131 MPa, 277 MPa and 141 MPa, respectively. Comparing with the Cr-free steel strip, the annealing treatment has a greater effect on the yield strength of the Cr-containing steel strips. In addition, after annealing, the ultimate tensile strength of the Cr-containing steel strips is higher than that of the Cr-free steel strip. The

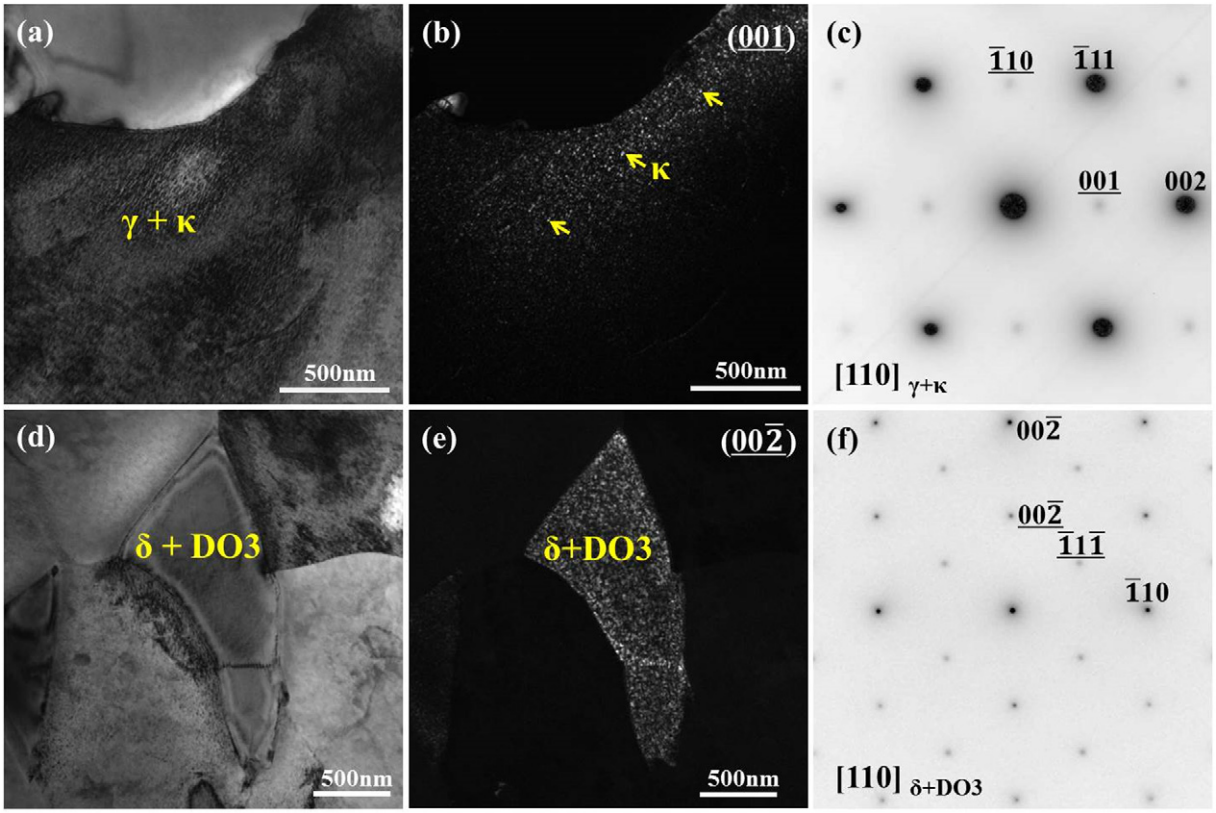


Fig. 4. TEM micrographs of γ-austenite (a–c) and δ-ferrite (d–f) in the as-cast 0Cr steel strip: (a) BF image of γ-austenite; (b) DF from $(001)_\kappa$; (c) the SADP along $[110]_\gamma$ zone axis from the γ-austenite (hkl: austenite; hkl: κ-carbide); (d) BF image of δ-ferrite; (e) DF from (00$\bar{2}$) superlattice reflection of the ordered phase; (f) the SADP along $[110]_\delta$ zone axis from the δ-ferrite (hkl: ferrite; hkl: DO3).

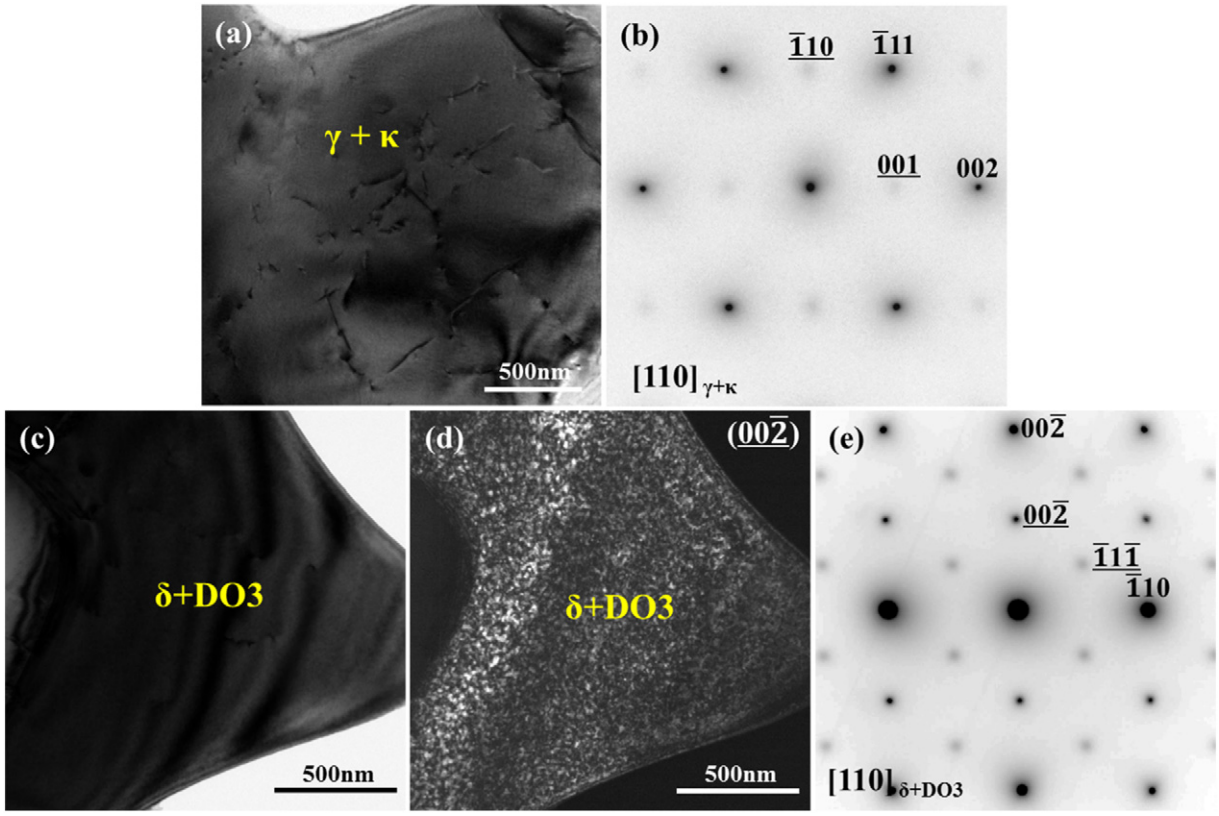


Fig. 5. TEM micrographs of γ-austenite (a–b) and δ-ferrite (c–e) in the as-cast 3Cr steel strip: (a) BF image of γ-austenite; (b) the SADP along $[110]_{\gamma}$ zone axis from the γ-austenite (hkl: austenite; hkl: κ-carbide); (c) BF image of δ-ferrite; (d) DF from (00$\bar{2}$) superlattice reflection of the ordered phase; (e) the SADP along $[110]_{\delta}$ zone axis from the δ-ferrite (hkl: ferrite; hkl: DO3).

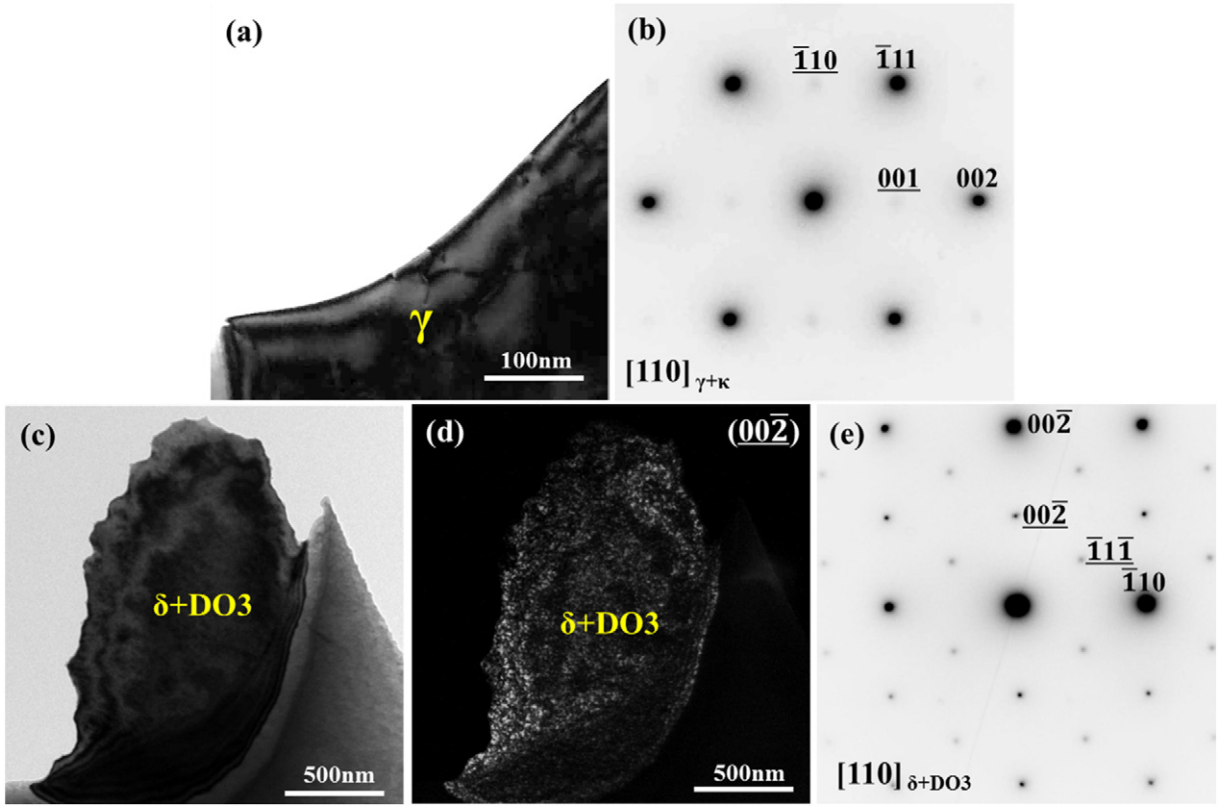


Fig. 6. TEM micrographs of γ-austenite (a–b) and δ-ferrite (c–e) in the as-cast 6Cr steel strip: (a) BF image of γ-austenite; (b) the SADP along $[011]_{\gamma}$ zone axis from the γ-austenite (hkl: austenite; hkl: κ-carbide); (c) BF image of γ-austenite; (d) DF from (00$\bar{2}$) superlattice reflection of the ordered phase; (e) the SADP along $[110]_{\delta}$ zone axis from the δ-ferrite (hkl: ferrite; hkl: DO3).

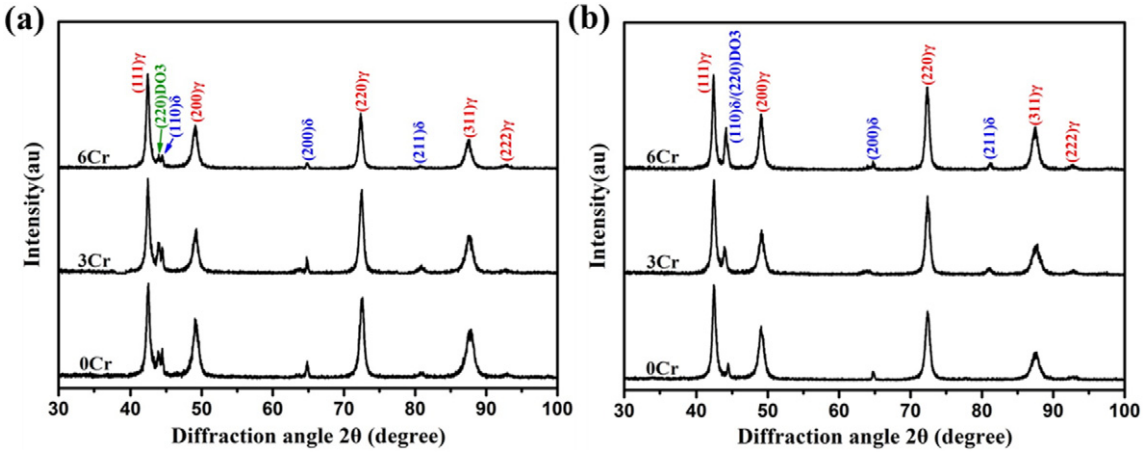


Fig. 7. X-ray diffraction patterns of the CR20% and CR20% + annealed steel strips.

highest product (38.0 GPa %) of ultimate tensile strength and total elongation (UTS × TE) is achieved in the as-cast 6Cr steel strip.

Fig. 3 (c) and (d) show the true stress-strain curves and strain hardening rate curves of the as-cast and annealed Fe-20Mn-9Al-1.2C-xCr steel strips. As shown in Fig. 3 (c), the addition of Cr significantly increases the strain hardening rate of the low-density steel strips. For the Cr-free steel, the strain hardening rate curve exhibits a 'hump', which is similar to the 'hump' associated with the deformation twinning occurred in a TWIP steel [37]. However, the hump of the steel strips becomes weak with Cr addition, and the strain hardening rate of the two as-cast Cr-containing steel strips gradually decreases during the deformation. As shown in Fig. 3 (d), the strain hardening rate of all the annealed steel strips is lower than that of the corresponding as-cast steel strips. A double hump with higher amplitude is observed in the strain hardening rate curve of the annealed Cr-free steel strip. A nearly stable platform is observed in the strain hardening rate curve of the annealed 3Cr steel strip, whereas the strain hardening rate is still monotonically decreased in the annealed 6Cr steel strip. It indicates that the Cr element has a significant effect on the deformation behavior of the annealed low-density steel strips.

3.1.3. Precipitated phase of steel strips

TEM was performed to elucidate the effect of Cr addition on the nano-scale precipitated phase in the low-density steel strips. TEM micrographs in Fig. 4 show the precipitated phases in γ-austenite and δ-ferrite of the as-cast Cr-free steel strip. Fig. 4 (a) is a bright-field (BF) image taken from the γ-austenite, in which some scattered spots are present in γ-austenite. Correspondingly, the dark-field (DF) image using (001) indicates that the nanoparticles are distributed in γ-austenite, as shown in Fig. 4 (b). Fig. 4 (c) and (f) are two selected area diffraction patterns (SADPs) taken from the γ-austenite and δ-ferrite along [110] zone axis, respectively. As shown in Fig. 4 (c), the superlattice reflection spots indicate that the ordered κ-carbides are present in the γ-austenite with the orientation relationship of $[110]_\gamma//[110]_\kappa$ [13,18]. The measured volume fraction and average size of κ-carbides in the γ-austenite are 3% and 6 nm, respectively. The measured lattice parameter of the κ-carbides and γ-austenite in the as-cast 0Cr steel strip is 0.387 nm and 0.375 nm, respectively. The ideal stoichiometry of the κ-carbide is $(Fe, Mn)_3AlC$, which has a perovskite crystal [5]. In this crystal structure, Fe or Mn atoms occupy the cornered positions, Al atoms occupy the position of the face center, and C is in the center of the cube

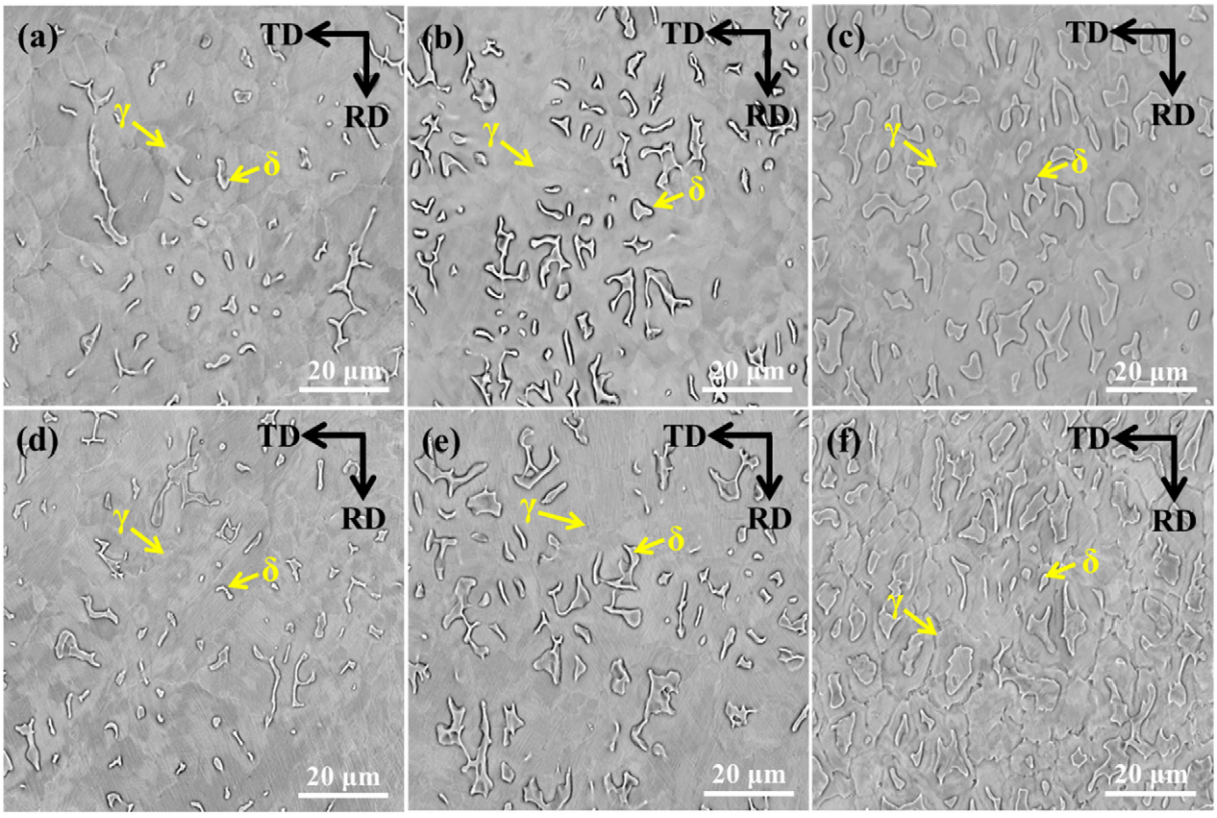


Fig. 8. SEM micrographs of the CR20% and CR20% + annealed steel strips (RD represents rolling direction and TD represents transverse direction). CR20%: (a) 0Cr, (b) 3Cr, (c) 6Cr; CR20% + annealed: (d) 0Cr, (e) 3Cr, (f) 6Cr.

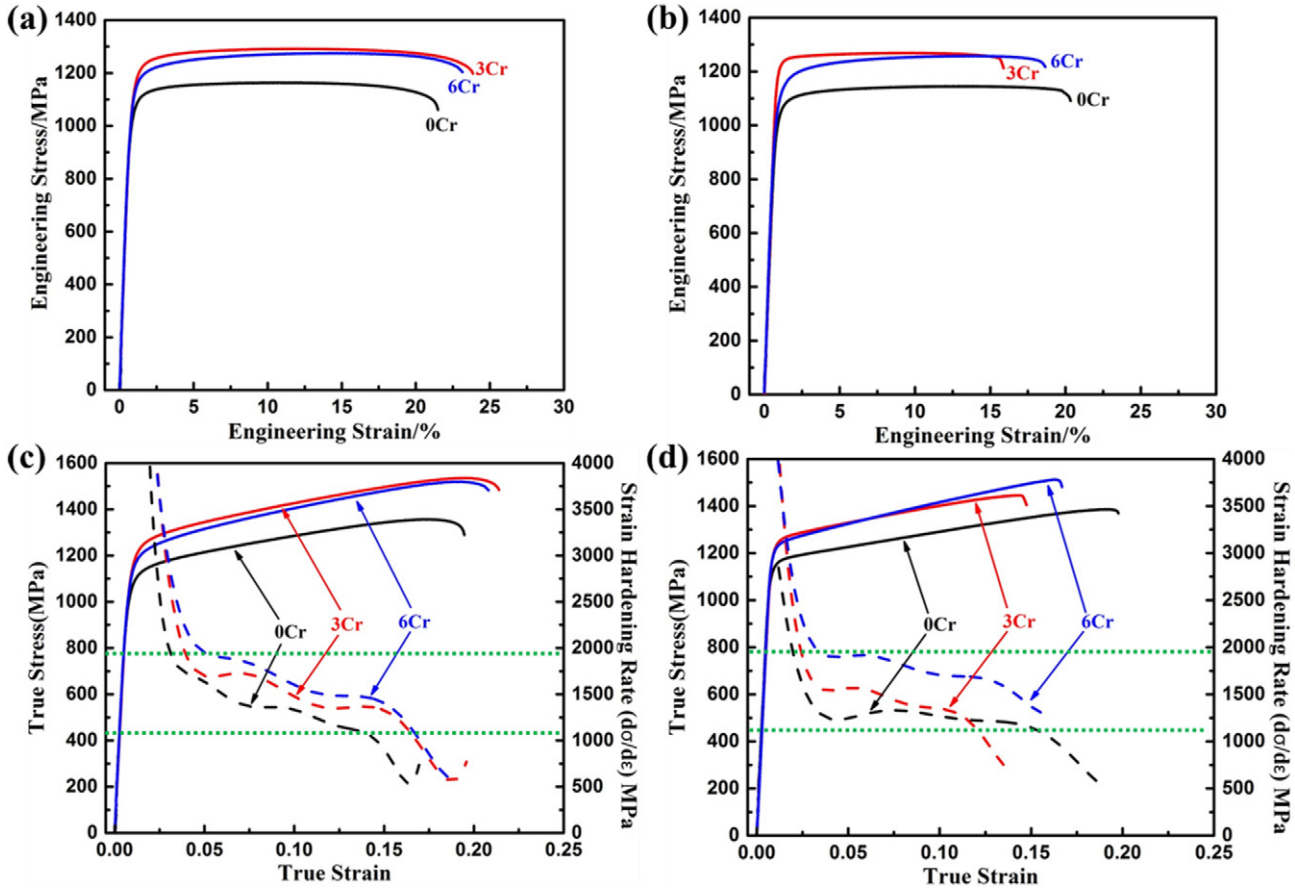


Fig. 9. Engineering stress-strain curves of the CR20% (a) and CR20% + annealed (b) steel strips; true stress-strain curves and strain hardening rate curves of the CR20% (c) and CR20% + annealed (d) steel strips.

crystal structure (½, ½, ½) [5]. As shown in Fig. 4 (d), the irregularly shaped δ-ferrite insets in γ-austenite, and the DF image using $(00\bar{2})$ shows clear ordering of the δ-ferrite, as shown in Fig. 4 (e). According to the diffuse superlattice reflection spots appeared in Fig. 4 (f), the ordered phase in the δ-ferrite is DO3 phase with size of 26 ± 3 nm [36,38,39]. For the SADP of δ-ferrite in the Fe-Mn-Al-C steels, when h, k and l (the Miller indices of the $\vec{g}$ vector) are unmixed [30,38,39], the reflections belong to DO3 phase. It means that in SADP of ferrite, DO3 reflection occurs where h, k and l (the Miller indices of the $\vec{g}$ vector) are all odd or even. Therefore, the δ-ferrite contains DO3 ordered phase. M.C. Ha et al. [39] and K.T. Park et al. [36] reported that some domains bounded by the swirled boundaries occurred in the DF image using (002) when B2 and DO3 coexisted in the δ-ferrite. However, the similar morphology does not occur in the DF images using $(00\bar{2})$, and it is confirmed that the precipitate in the ferrite is mainly DO3 phase.

Fig. 5 shows the microstructure of the as-cast 3Cr steel strip. Fig. 5 (a) is the BF image of γ-austenite, and Fig. 5 (b) is the SADP taken along [110] zone axis from the γ-austenite. The reflection intensity of the superlattice is so weak that the precipitation cannot be recognized in the DF image from (001) superlattice reflection. It indicates that the volume fraction of κ-carbide decreases compared with the Cr-free sample shown in Fig. 4 (b) [30,40]. The measured lattice parameter of the κ-carbides and γ-austenite in the as-cast 3Cr steel strip is 0.387 nm and 0.376 nm, respectively. According to Fig. 5 (d), a DF image using $(00\bar{2})$ shows the DO3 nanoparticles with size of 30 ± 3 nm are densely distributed in the δ-ferrite. It is consistent with the superlattice reflections in the SADP along [110] in the δ-ferrite [36,38], as shown in Fig. 5(e).

Fig. 6 is TEM micrographs of the as-cast 6Cr steel strip. Fig. 6 (a) is the BF image of the γ-austenite. Fig. 6 (b) shows SADP taken along [110] zone axis from the γ-austenite, the reflection intensity of the superlattice is barely visible, which indicates that the volume fraction of κ-carbide is so little. The measured lattice parameter of the κ-carbides and γ-austenite in the as-cast 6Cr steel strip is 0.389 nm and 0.384 nm, respectively. According to the DF image using $(00\bar{2})$ and the SADP along [110] in the δ-ferrite shown in Fig. 6

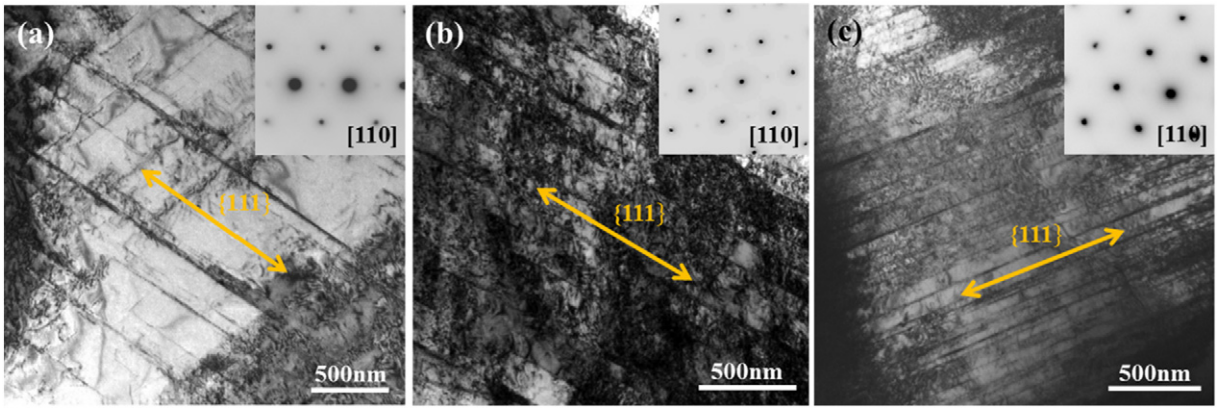


Fig. 10. TEM micrographs show the microstructures of γ-austenite in the 0Cr (a), 3Cr (b) and 6Cr (c) steel strips after cold rolling with 20% reduction.

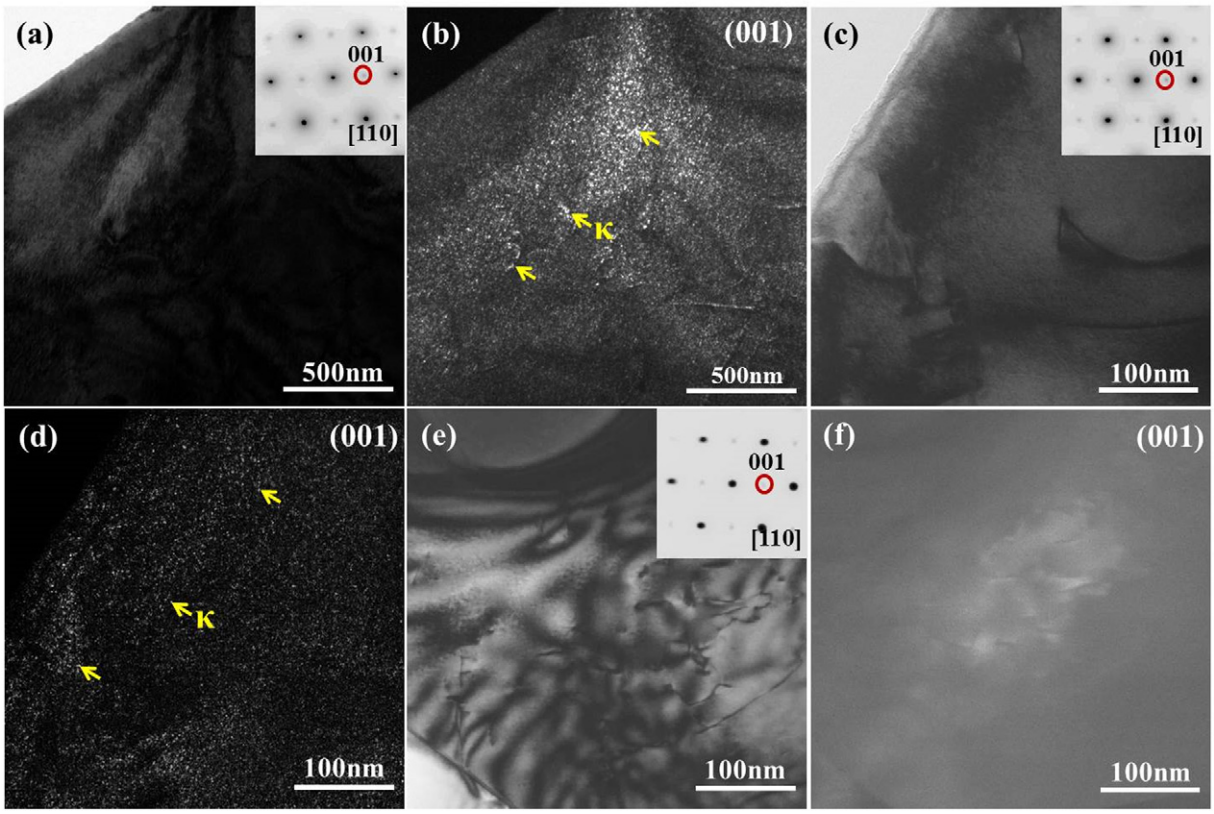


Fig. 11. TEM micrographs of γ-austenite in the annealed steel strips: (a) BF image of γ-austenite in the 0Cr steel strip; (b) DF from (001) superlattice reflection of the κ-carbides in the 0Cr steel strip; (c) BF image of γ-austenite in the 3Cr steel strip; (d) DF from (001) superlattice reflection of the κ-carbides in the 3Cr steel strip; (e) BF image of γ-austenite in the 6Cr steel strip; (f) DF from (001) superlattice reflection of the κ-carbides in the 6Cr steel strip.

(d) and (e), DO3 phase with size of 28 ± 4 nm precipitates in the δ-ferrite. Comparing with the DO3 phase precipitated in the 3Cr steel strip shown in Fig. 5 (d), there is no significant change in the size of the DO3 phase in the 6Cr steel strip.

3.2. Microstructures and properties of cold-rolled steel strips

3.2.1. Microstructures of the cold-rolled steel strips

Fig. 7 is the X-ray diffractogram of the CR20% (cold rolled with 20% reduction) and CR20% + annealed (further annealed at 600 °C for 3 h after cold rolling) steel strips. The constituent phases are still γ-austenite and δ-ferrite with ordered DO3 phase after cold rolling, which is completely consistent with the as-cast steel strips. After further annealing, the change from double-peak to single-peak between 43.5° and 45.0° also occurs in the CR20% + annealed steel strips.

Fig. 8 is the SEM micrographs of the CR20% and CR20% + annealed steel strips. There is no significant change in the morphology and volume fraction of the δ-ferrite in the steel strips after cold rolling. The volume fraction of δ-ferrite in the cold-rolled steel strips is 5%, 12% and 18%, respectively. The addition of Cr results in an increase of δ-ferrite fraction in the cold-rolled steel strips. Further annealing at 600 °C has no significant effect on the microstructure of the cold-rolled steel strips.

3.2.2. Mechanical properties of the cold-rolled steel strips

Fig. 9 (a) and (b) show the engineering stress-strain curves of the cold-rolled steel strips. After cold rolling, the strength of Cr-containing steel strips exceeds that of Cr-free steel strip, which is in contrast to the as-cast steel strips. And, the ultimate tensile strength is 1154 MPa, 1270 MPa and 1264 MPa for the CR20% 0Cr, 3Cr and 6Cr steel strips, respectively. The yield strength also increases with Cr addition. Compared with the CR20% steel strips, the yield strength and the ultimate tensile strength of the CR20% + annealed steel strips do not change significantly.

As shown in Fig. 9 (c), the strain hardening rates of the CR20% steel strips decrease gradually with increasing strain. The addition of Cr also increases the strain hardening rate of the CR20% steel strips. Comparing with the CR20% steel strips, annealing makes the strain hardening rates of the CR20% + annealed 0Cr and 3Cr steel strips clearly decrease at the early stage.

To elucidate the reason for the difference in reinforcement effect of the cold-rolled steel strips, TEM was used to examine the microstructures of CR20% 0Cr, 3Cr and 6Cr steel strips. The BF images and the SADP taken from [110] zone axis of the γ-austenite in the three CR20% steel strips are shown in Fig. 10. According to these TEM micrographs, {111} planar slip bands are present in the austenite of the CR20% steel strips. Furthermore, there is a significant difference in the slip band spacing between the Cr-free and Cr-containing steel strips. The slip band spacing was measured at least three different regions in each specimen. The measured results show that the average spacing of the slip bands in 0Cr, 3Cr and 6Cr steel strips is 371 ± 47 nm, 100 ± 15 nm and 45 ± 13 nm, respectively. The difference in the slip band spacing should be responsible for the difference in reinforcement effect of the steel strips after cold rolling. The smaller slip band spacing, the stronger reinforcement effect after cold rolling.

4. Discussions

4.1. Effect of Cr on strength

1) As-cast steel strips

The precipitation of nano-scale κ-carbides in the low-density steel can effectively enhance the strength of steel [13,37]. In this work, the volume fraction of the κ-carbides decreases with Cr addition, as shown in Figs. 4–6. Thus, the Cr addition makes the yield strength decrease. And, the precipitation strengthening is low in both the as-cast 3Cr and 6Cr steel strips due to the small volume fraction of κ-carbide. As a result, the yield strength of the as-cast 6Cr steel strip is higher than that of the as-cast 3Cr steel strip, due to the solid solution strengthening caused by high Cr content.

The TEM micrographs of the γ-austenite in the annealed steel strips are shown in Fig. 11. The superlattice reflection spots appear in the SADP along [110] zone axis taken from the γ-austenite. And the DF images using (001) indicate that a large number of nano-scale κ-carbides in the annealed 0Cr and 3Cr steel strips. The measured volume fraction and average size of the κ-carbides in the annealed 0Cr steel strip reach 7% and 8 nm, respectively. And, in the annealed 3Cr steel strip, they are 4% and 4 nm, respectively. It can be concluded that, annealing treatment at 600 °C promotes the precipitation and growth of κ-carbides, which leads to an increase in the yield strength, as shown in Fig. 3. According to Fig. 11, the Cr addition makes the size and number of κ-

carbides decrease significantly in the annealed steel strips. The reason should be that addition of Cr results in expanding the solid solubility of Al and C in the Fe-Mn-Al-C steel [28,29], which reduces the driving force of the formation of κ-carbide for a given Al and C content. Therefore, the higher the Cr content, the less the κ-carbide precipitation. And the yield strength of the annealed Cr-containing steel strips also decreases.

2) Cold-rolled steel strips

After cold rolling with 20% reduction, the strength of the Cr-containing steel strips is higher than that of Cr-free steel strips, as shown in Fig. 9. E. Welsch et al. [11] reported a clear correspondence between the slip band spacing and the stress required for deformation. The smaller the slip band spacing, the greater the stress required for deformation. In this work, there is a higher density of the slip bands in the Cr-containing cold-rolled steel strips than that of the Cr-free cold-rolled steel strips. The high-density slip bands significantly block the movement of dislocations and increase the deformation difficulty, which leads to an increase in strength of the Cr-containing steel strips. Thus, high density of the slip bands is responsible for the high strength of the Cr-containing cold-rolled steel strips.

The BF images of the γ-austenite in the CR20% + annealed 0Cr, 3Cr and 6Cr steel strips are shown in Fig. 12 (a), (c) and (e), respectively. The SADPs taken along [110] and [100] zone axis from the γ-austenite are shown in the inset of the BF images. It indicates that the annealing treatment at 600 °C also promotes the precipitation of κ-carbides in the CR20% steel strips. The DF images of κ-carbides using (001) shown in Fig. 12 (b) and (d) indicate that the κ-carbides are mainly precipitated at the dislocation entanglement in the CR20% + annealed 0Cr and 3Cr steel strips. But in the CR20% + annealed 6Cr steel strip, the size of κ-carbides is too small to observe in the DF image, as shown in Fig. 12 (f). Although annealing promotes the precipitation of κ-carbides, the yield strength of the CR20% + annealed steel strips does not significantly increase, as shown in Fig. 9. This can be attributed to the fact that the high density slip bands become the main strengthening factor for the cold-rolled steel strips.

4.2. Effect of Cr on strain hardening

The addition of Cr results in an obvious difference in the strain hardening rate curves of the annealed steel strips. In order to elucidate the effect of Cr on the deformation behavior of the annealed steel strips, additional TEM observation was made to the annealed 0Cr and 6Cr steel strips at the early (true strain of 0.075) and final strain stages (tensile fracture).

1) At early strain stage

The BF images of the annealed 0Cr and 6Cr steel strips at a true strain of 0.075 are shown in Fig. 13. At this strain stage, some slip bands are present in the austenite. As shown in Fig. 13 (a) and (c), the addition of 6Cr makes the spacing of the slip bands smaller than that of 0Cr steel strips at this strain stage. The higher density of slip bands caused by the same strain would result in higher strain hardening rate in the 6Cr steel strips, because it makes the dislocations glide more difficult [14,41,42]. A typical dislocation network structure is also present in the austenite of the 0Cr steel strip, as shown in Fig. 13 (b), which is similar to the microstructure in austenite-ferrite Fe-18.1Mn-9.6Al-0.65C steel at a true strain of 0.06 [9]. In addition, the dislocations between the slip bands become entangled and agglomerated in some regions of the 6Cr steel strip, as shown in Fig. 13 (d).

The slip bands and the dislocation arrays shown in Fig. 13 (a) and (c) indicate a typical planar slip characteristic [9], which is related to the

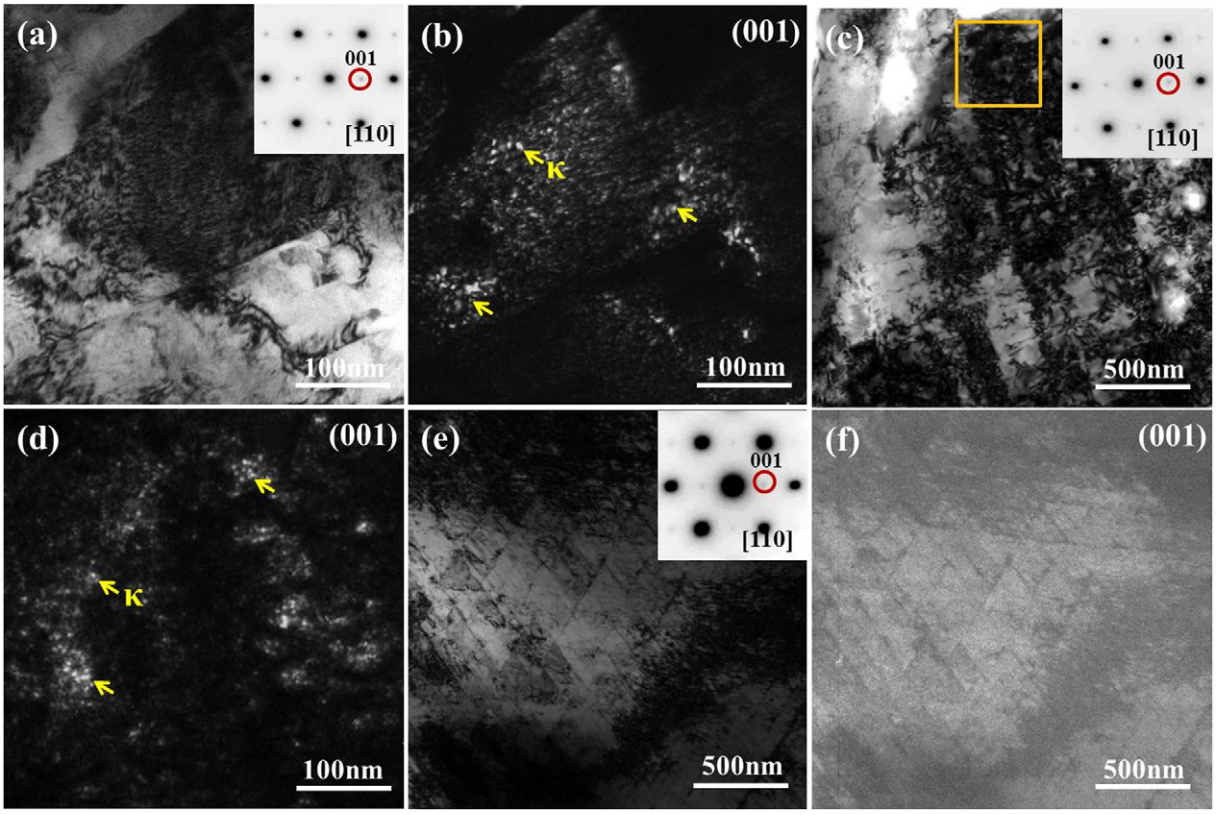


Fig. 12. TEM micrographs of γ-austenite in the CR20% + annealed steel strips: (a) BF image of γ-austenite in the 0Cr steel strip; (b) DF from (001) superlattice reflection of the κ-carbides in the 0Cr steel strip; (c) BF image of γ-austenite in the 3Cr steel strip; (d) DF from (001) superlattice reflection of the κ-carbides in the 3Cr steel strip; (e) BF image of γ-austenite in the 6Cr steel strip; (f) DF from (001) superlattice reflection of the κ-carbides in the 6Cr steel strip.

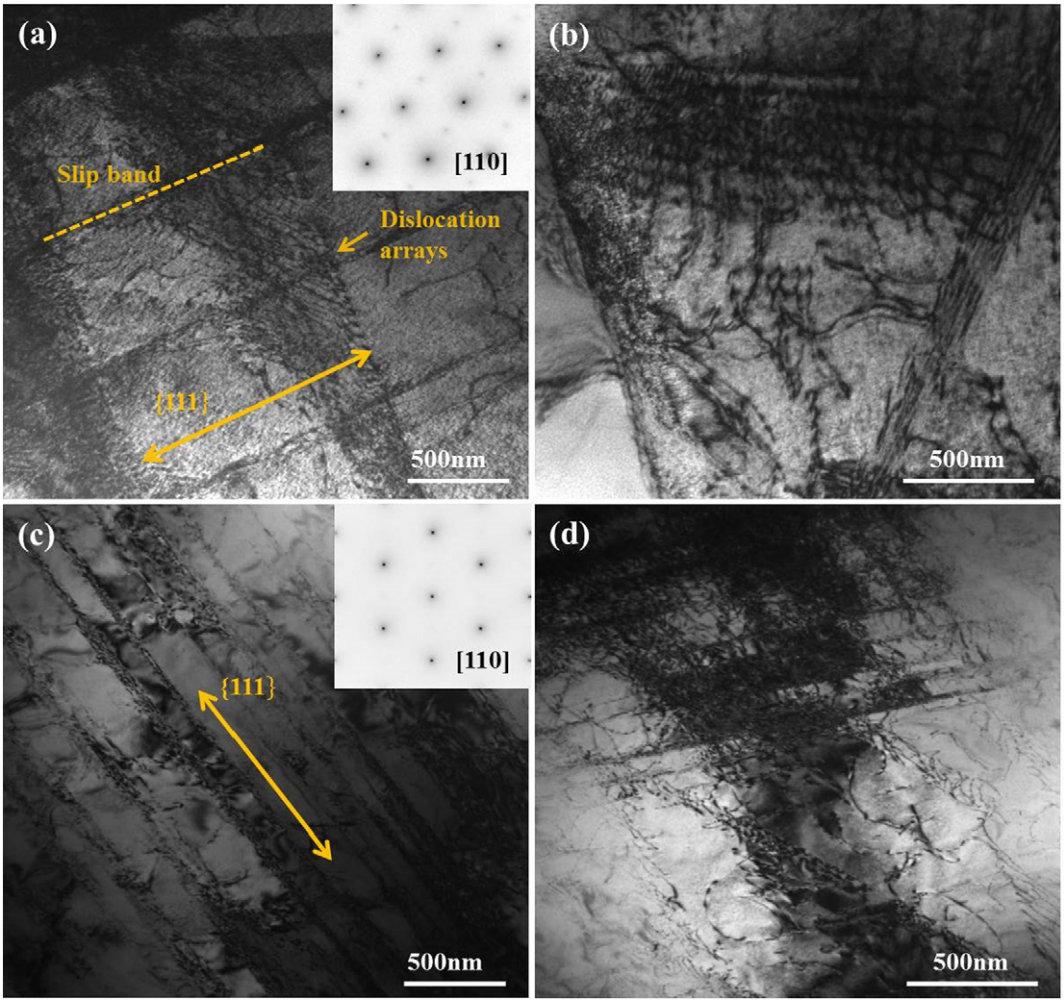


Fig. 13. TEM micrographs of the annealed 0Cr and 6Cr steel strips at a true strain of 0.075 during tensile deformation: (a) slip bands in 0Cr steel strips, (b) dislocation network structure in 0Cr steel strip, (c) slip bands in 6Cr steel strips, (d) scattered dislocations in 6Cr steel strips.

interaction between nano-scale κ-carbides and dislocations [11,13]. The nano-scale κ-carbides act as reinforcing phases to hinder the glide of dislocations by providing energy barrier. The decrease of the κ-carbides caused by Cr addition contributes to the formation of dislocations and increases the density of the slip bands. Therefore, the density of the slip bands in the Cr-containing steel strips is higher at the early strain stage.

Actually, when the first dislocation meets with the barrier of κ-carbides, the following dislocations activated on the same plane are piled up to assist the first dislocation to overcome the hindrance of κ-carbides. The distorted κ-carbides with the order in the short range are hard to be recovered, resulting in a glide plane softening [3,10,14,39]. Glide plane softening makes dislocation glide easier. Therefore, the effect of κ-carbide on the strain hardening is mainly concentrated at the early strain stage. More precipitation of the κ-carbides in the annealed 0Cr steel strips supplies more obstruction to the dislocation glide and generation of the slip bands, which results in a low strain hardening rate.

It is should be noted that the annealing treatment after cold rolling also promotes the precipitation of κ-carbides, but the effect of the newly formed κ-carbides on the strain hardening rate (Fig. 3 (c) and (d)) is weaker than that of the as-cast steel strips (Fig. 9 (c) and (d)). After cold rolling, there are a large number of dislocations and slip bands in the steel strips, as shown in Fig. 10. The annealing treatment after cold rolling promotes the precipitation of κ-carbides, but does not eliminate these dislocations and slip bands, as shown in Fig. 12. At early strain stage, these deformation microstructures result in high yield stress of the steel strips, which makes dislocations easier to shear κ-carbides.

2) At final strain stage

With the strain progressing, the dislocations are constantly generated. Due to the glide plane softening, the distorted κ-carbides are continuously sheared by a large number of dislocations, which further reduce the pinning effect of κ-carbides [3,10,14]. Therefore, the effect of κ-carbides as an energy barrier becomes increasingly weaker. And at final strain stage of the 0Cr and 6Cr steel strips, the difference in strain hardening rate caused by κ-carbides becomes small, as shown in Fig. 3.

Fig. 14 shows the BF images of the annealed 0Cr and 6Cr steel strips after tensile fracture (The true strains of the 0Cr and 6Cr steel strips are 0.255 and 0.298, respectively.). The microstructures of two steel strips are similar after tensile fracture, which can be confirmed that effect of κ-carbides is very weak at final strain stage. Moreover, besides a large number of slip bands, many microbands (marked with white arrows) also appear in the austenite matrix of 0Cr and 6Cr steel strips, as shown in Fig. 14 (b) and (d). The microbands consisting of a large number of dislocations would also result in strain hardening. Therefore, another hardening mechanism caused by the well-developed microbands appears at final strain stage of annealed 0Cr and 6Cr steel strips.

In addition, the dislocation entanglements occur at the phase boundary between ferrite and austenite, which can be attributed to the inconsistent deformation of ferrite and austenite during the deformation process. And the slip bands can extend through a grain boundary, and lead to many small steps at the grain boundary, as shown in Fig. 14 (c). The SADPs along $[110]_{\gamma}$ zone axis on both sides of the grain

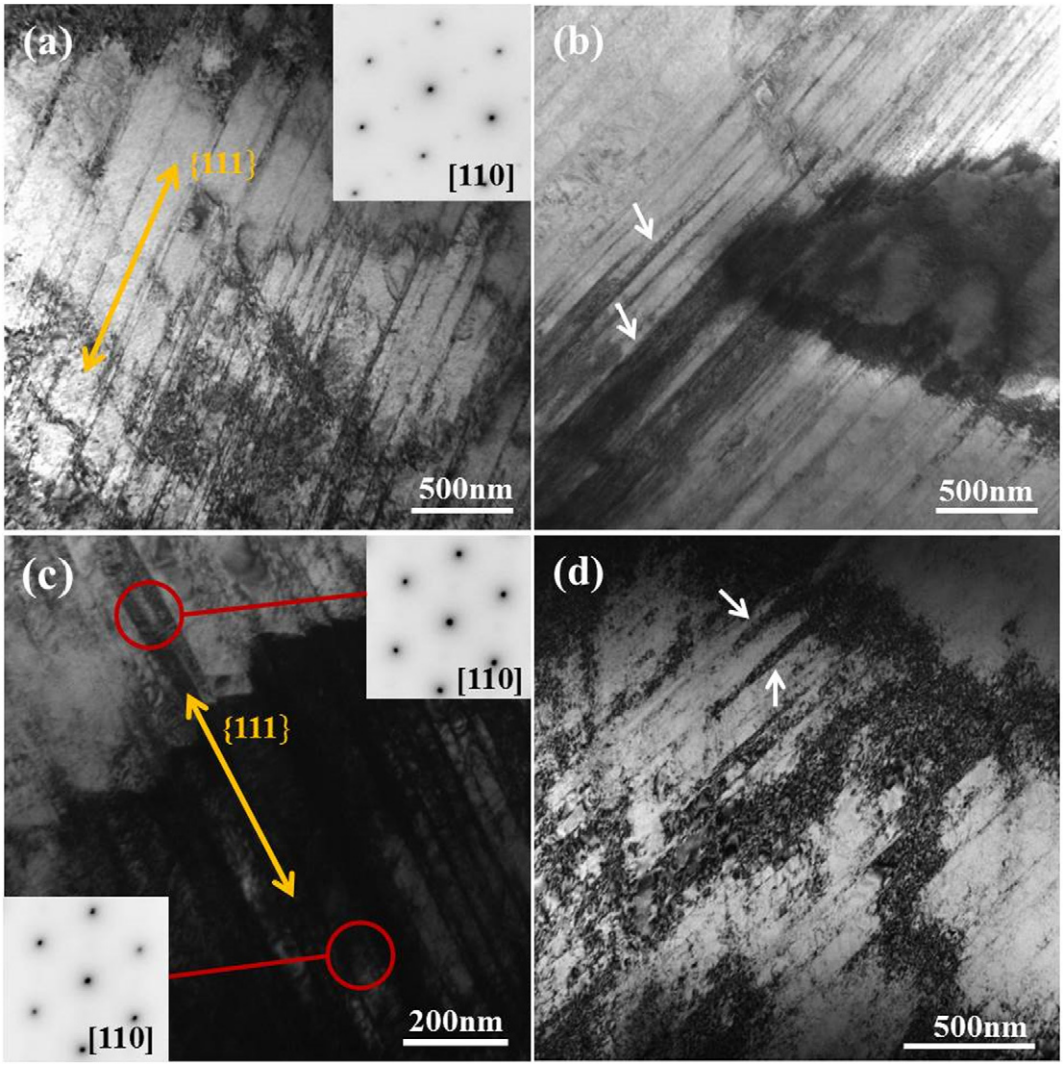


Fig. 14. TEM micrographs of the annealed 0Cr and 6Cr steel strips after tensile fracture: (a) slip bands in 0Cr steel strips, (b) intersections between ferrite and microbands in 0Cr steel strips, (c) shearable grain boundary in 6Cr steel strips, (d) microbands in 6Cr steel strips.

boundary indicate that this grain boundary is a low-angle grain boundary with a misorientation of 2.56°, as shown in Fig. 14 (c). The phenomenon is observed only at high strain and is rarely observed, which the reason is not clear.

3) Hardening mechanism

J.D. Yoo and K.T. Park [14] proposed that microbands result in continuous strain hardening of the fully austenitic Fe-28Mn-9Al-0.8C steel over 80 GPa·%. In contrast to the J.D. Yoo and K.T. Park, E. Welsch et al. [11] found that the constant high strain hardening rate of the Fe-30.4Mn-8Al-1.2C steel was attributed to the dynamic slip band refinement. However, it is found there are both slip bands and microbands in the deforming process of all the studied steel strips. And, formation of slip bands usually occurs at the early strain stage, and the microbands appear at the final strain stage. So a double hardening by slip bands and microbands acts at the different strain stages of this low-density steel. And at the final strain stage, the strain hardening rate caused by microbands is usually relatively low.

With respect to the reason for that the strain hardening rate of annealed 0Cr steel strip shows a clear double hump like feature shown in Fig. 3 (d). According to the above analysis, the first hump of the strain hardening rate is related to the slip-band refinement, and the second is due to the formation of microbands. After the beginning of plastic deformation, the generation of slip bands leads to the first hump of the strain hardening rate. As the slip band spacing reaches the critical value, the strain hardening rate decreases. Under high strain condition, the generation of microbands results in the second hump of the strain hardening rate. As strain increases further, the necking occurs and then specimen fractures by plastic instability. It is emphasized that the number of nano-scale κ-carbides plays a crucial role in the strain hardening rate at the early strain stage as discussed above. When the number of κ-carbides is great, the early strain hardening rate caused by slip bands is low and similar to that caused by microband hardening level. This makes the strain hardening rate of the anneal 0Cr steel strip show a clear double hump like feature as shown in Fig. 3 (d). But when number of κ-carbides is small as the Cr-containing steel strips, the early strain hardening rate caused by slip bands is obviously higher than that caused by microband hardening, which presents a gradually decrease trend in whole tensile process and the hump like feature becomes weaker.

5. Conclusions

The effect of the addition of Cr on the microstructure, mechanical properties and deformation behavior of low-density Fe-20Mn-9Al-1.2C steel strips prepared under near-rapid solidification has been

investigated. According to the obtained results, following conclusions can be drawn.

(1) The constituent phases in the low-density Fe-20Mn-9Al-1.2C-(0, 3, 6)Cr steel strips are mainly austenite and ferrite, and the ferrite with irregular shape is evenly distributed in the austenite matrix. The volume fraction and size of ferrite are significantly increased with the addition of Cr.
(2) There are some nano-scale κ-carbides and ordered DO3 phases in the austenite and ferrite of the Fe-20Mn-9Al-1.2C steel strips, respectively. With increasing the content of Cr, the volume fraction of nano-scale κ-carbides decreases.
(3) Due to the decrease of the nano-scale κ-carbides, the Cr addition decreases the yield strength of the as-cast and annealed steel strips. However, after cold rolling with 20% reduction, the strength of Cr-containing steel strips is higher than that of Cr-free steel strip. The ultimate tensile strength of CR20% 3Cr and 6Cr steel strips reaches 1270 MPa and 1264 MPa, respectively, and the elongation remains about 21%.
(4) The decrease of the κ-carbide caused by Cr addition could promote the formation of slip bands at the early strain stage. It results in the high-density slip bands during deformation and thereby a higher strain hardening rate of Cr-containing steel strips.
(5) A double (Slip-band and Microband) hardening mechanism is proposed in the strain hardening rate of the low-density steel strips. The hardening of slip bands and microbands acts at the early and final stages of the deformation process, respectively.

Declaration of interest statement

Declarations of interest: none.

Data availability statement

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

CRediT authorship contribution statement

Jianlei Zhang: Writing - original draft, Methodology, Investigation, Writing - review & editing. **Conghui Hu:** Investigation, Writing - review & editing. **Yunhu Zhang:** Investigation, Supervision. **Jiehua Li:** Investigation. **Changjiang Song:** Investigation, Supervision, Funding acquisition, Project administration, Writing - review & editing. **Qijie Zhai:** Supervision, Funding acquisition, Project administration.

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